

BJORNOYA, Svalbard and Jan Mayen

WMO: 010280

Lat: 74.504N

Lon: 19.001E

Elev: 18

StdP: 101.11

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-20.3	-17.6	-22.6	0.5	-20.1	-19.8	0.7	-17.4	20.7	-2.2	18.6	-2.2	7.7	40	1.023

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.7	10.6	9.4	9.5	8.5	8.6	7.8	9.6	10.4	8.7	9.3	7.9	8.5	7.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	7.1	10.1	8.2	6.7	9.1	7.4	6.4	8.3	28.5	10.5	26.4	9.4	24.7	8.5	16.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	15.4	13.9	DB	-19.2	14.2	4.6	2.6	-22.5	16.0	-25.2	17.6	-27.8	19.0	-31.1	20.9	(4)
(5)			WB	-19.5	11.7	4.4	1.7	-22.7	12.9	-25.3	13.9	-27.8	14.8	-31.1	16.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.4	-4.5	-5.0	-5.4	-3.6	-0.1	3.0	5.4	5.6	4.1	0.9	-1.7	-3.5
	DBStd	5.39	5.32	5.00	4.94	4.20	2.82	2.12	2.10	1.99	2.22	2.79	3.52	4.72
	HDD10.0	3793	450	419	478	407	314	211	145	138	178	283	350	419
	HDD18.3	6833	708	652	737	657	573	461	402	396	428	542	600	677
	CDD10.0	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.0	8.2	8.5	7.9	7.1	6.3	5.4	5.7	5.3	6.6	7.3	7.8	8.2
Precipitation	PrecAvg	403	38	37	34	27	21	22	28	33	46	42	38	39
	PrecMax	529	125	111	79	55	51	91	83	86	95	104	74	95
	PrecMin	263	11	5	8	4	4	4	2	7	10	12	6	7
	PrecStd	66	21	20	16	12	11	15	16	20	20	19	15	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	3.3	3.1	4.6	6.7	10.2	13.5	12.3	10.9	8.0	5.2	4.0
		MCWB	2.9	2.4	2.2	3.6	5.4	8.3	11.5	10.9	9.7	7.1	4.4	3.1
	2%	DB	2.8	2.5	2.4	3.1	5.3	8.0	10.8	10.5	9.2	6.7	4.2	3.1
		MCWB	2.1	1.7	1.6	2.3	4.3	6.8	9.5	9.4	8.4	6.1	3.6	2.4
	5%	DB	2.1	1.8	1.7	2.4	4.3	6.8	9.3	9.5	8.1	5.8	3.2	2.4
		MCWB	1.4	1.1	1.0	1.7	3.4	5.7	8.4	8.7	7.5	5.2	2.7	1.7
	10%	DB	1.2	0.9	1.0	1.7	3.4	5.9	8.4	8.6	7.3	4.7	2.5	1.7
		MCWB	0.6	0.2	0.4	1.0	2.5	4.9	7.6	7.9	6.8	4.2	1.8	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.7	2.5	3.8	5.7	8.7	11.8	11.1	9.9	7.3	4.6	3.4
		MCDB	3.6	3.2	2.8	4.4	6.3	9.6	13.1	12.2	10.6	7.7	5.0	3.8
	2%	WB	2.2	1.9	1.7	2.5	4.5	7.0	9.6	9.6	8.5	6.2	3.7	2.6
		MCDB	2.7	2.4	2.2	2.9	5.2	7.9	10.6	10.3	9.0	6.7	4.1	2.9
	5%	WB	1.5	1.2	1.1	1.8	3.6	5.9	8.6	8.8	7.7	5.3	2.8	1.8
		MCDB	2.0	1.8	1.6	2.2	4.2	6.6	9.2	9.3	8.1	5.8	3.2	2.3
	10%	WB	0.6	0.2	0.4	1.1	2.6	5.1	7.7	8.0	6.9	4.2	2.0	1.1
		MCDB	1.1	0.8	0.9	1.6	3.2	5.8	8.3	8.5	7.3	4.7	2.4	1.6
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.2	4.3	3.9	2.8	2.6	2.9	2.7	2.5	2.6	3.3	3.9
		MCDBR	3.7	4.1	4.0	3.3	3.6	3.8	4.9	4.2	3.6	3.1	3.1	3.6
		MCWBR	3.7	4.2	4.1	3.2	3.2	3.3	3.9	3.6	3.3	3.2	3.2	3.7
	5% WB	MCDBR	3.8	4.0	4.2	3.3	3.6	3.6	4.8	4.1	3.6	3.2	3.2	3.4
		MCWBR	3.9	4.1	4.2	3.3	3.3	3.3	4.0	3.7	3.3	3.2	3.3	3.5
Clear-Sky Solar Irradiance	taub	N/A	0.385	0.308	0.330	0.337	0.319	0.329	0.321	0.304	0.359	N/A	N/A	
	taud	N/A	2.390	2.153	2.321	2.420	2.540	2.539	2.538	2.447	2.586	N/A	N/A	
	Ebn at Noon	0	183	586	760	819	858	830	781	637	223	0	0	
	Edh at Noon	0	23	63	82	87	80	77	67	49	19	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.07	0.79	2.17	3.54	4.29	3.41	2.36	1.01	0.20	0.00	0.00	
	RadStd	0.00	0.01	0.09	0.26	0.28	0.37	0.30	0.23	0.11	0.03	0.00	0.00	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.18	+4.34	+3.43	N/A	N/A	N/A	-479	-481	N/A	N/A	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page